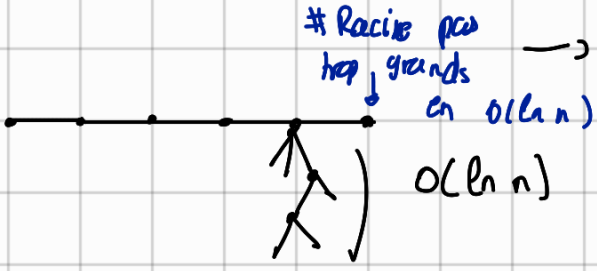
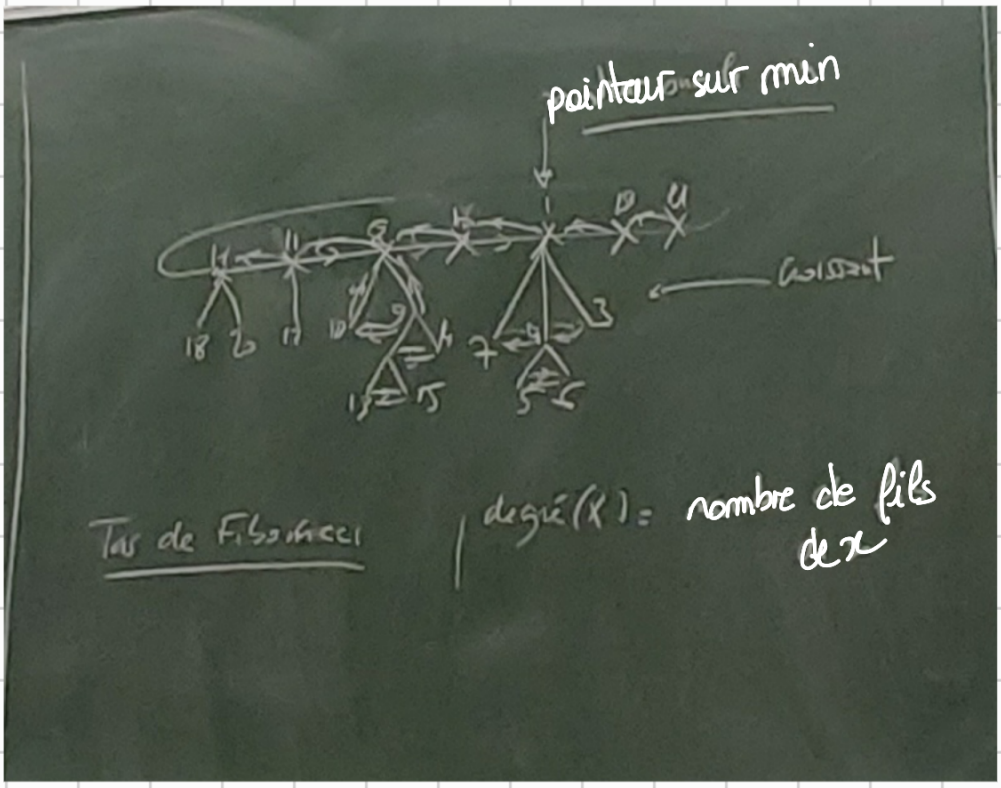
Tas de Fibonacci



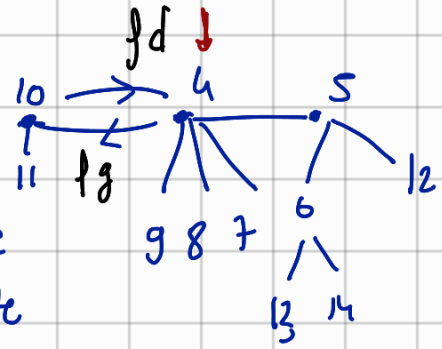
→ problème des tas binomiaux : structure trop rigide.

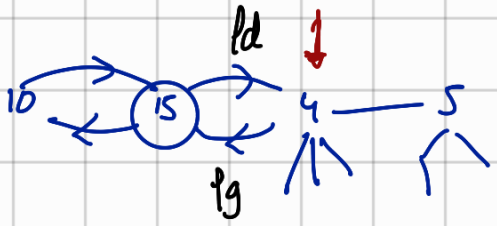
Dans les tas fibo : tout est doublement chainées.



insertion : on l'insère à gauche du min.

insérer 15





coût réel : constante

coût amorti : constante

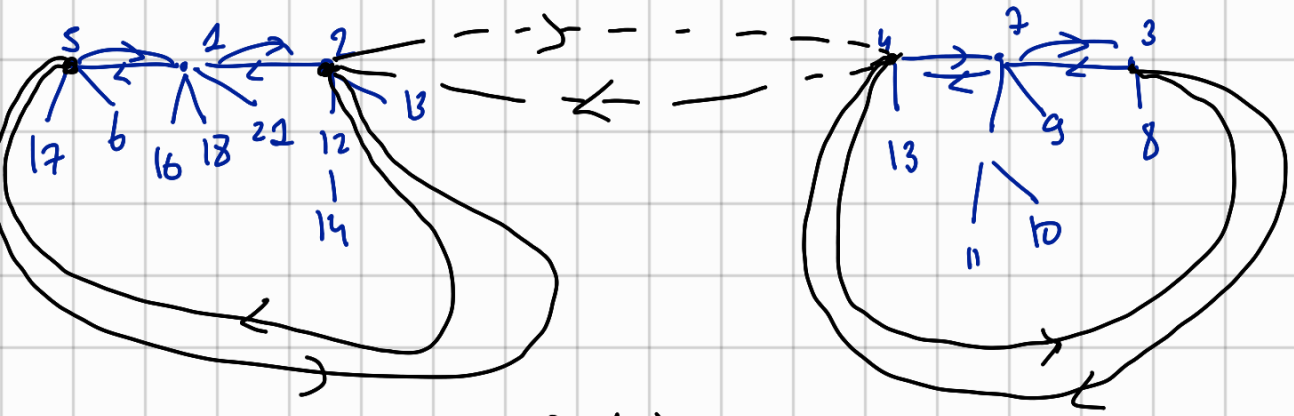
$+ \Delta\phi$ = constante

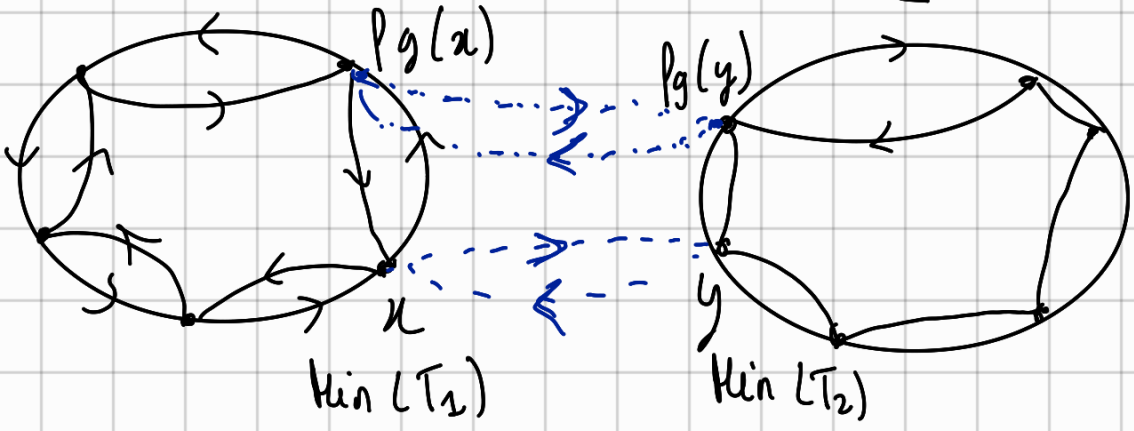
1 $+ 1$

Pour l'analyse Amortie :

$\Phi(T)$ = # de racines + 2 # de marge.

union : de 2 tas Fibonacci.





Min (T_1) = x

Min (T_2) = y

$f_g(x) \leftarrow y$

$fd(f_g(x)) \leftarrow fg(y)$

$f_g(y) \leftarrow x$

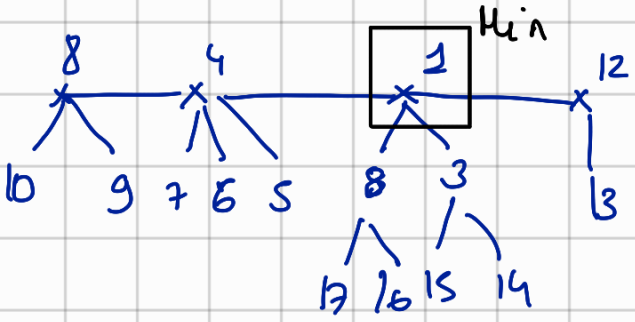
$fd(f_g(y)) \leftarrow f_g(x)$

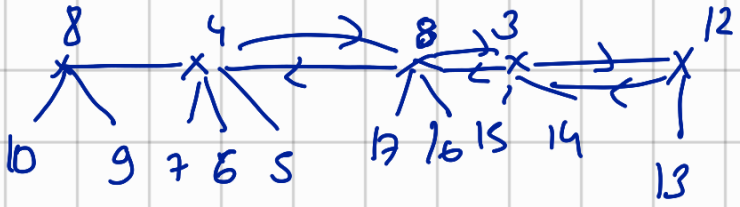
extraire le min

(on range la structure uniquement quand on extrait le min).

↳ extraire min.

↳ consolidation (ranger)

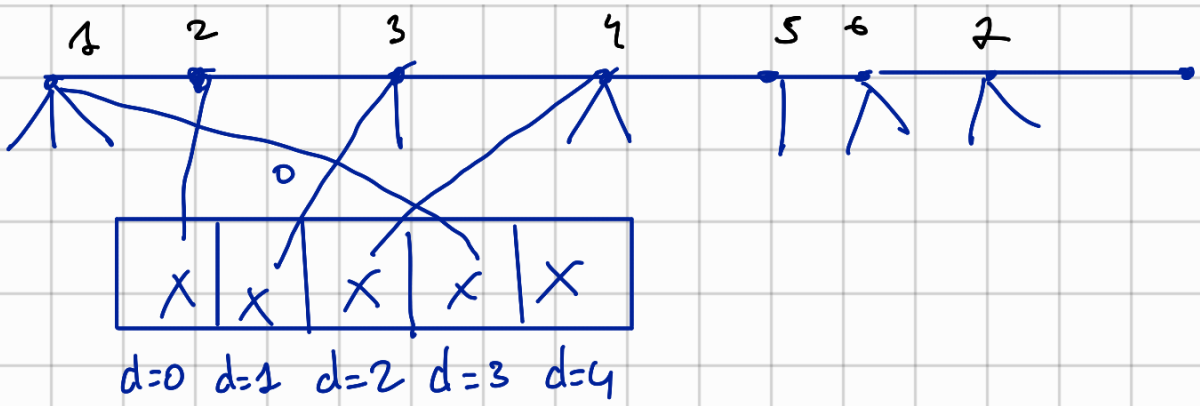


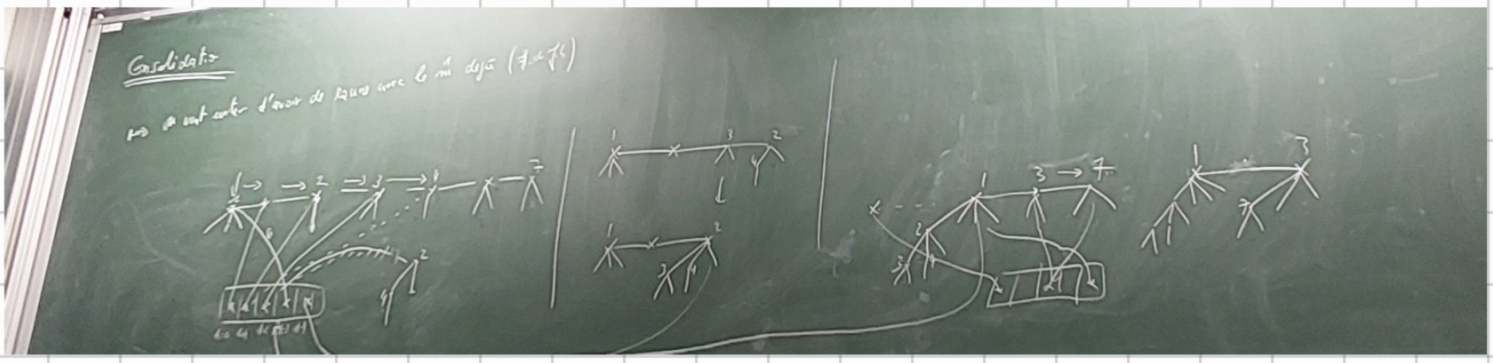


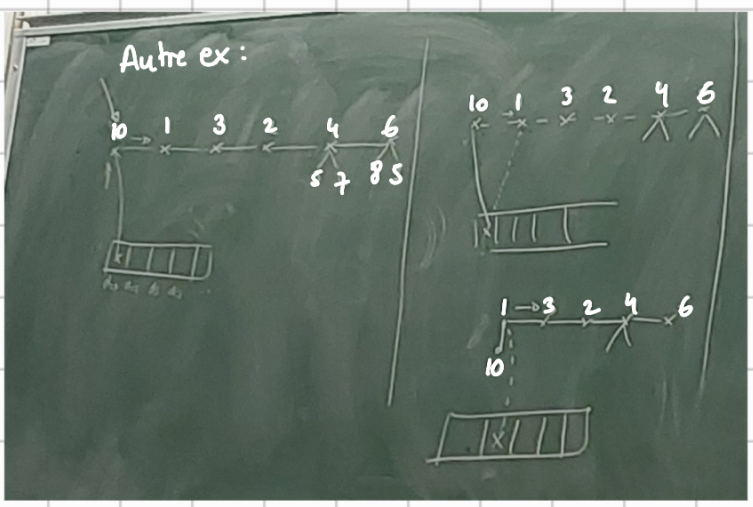
coût réel : constant

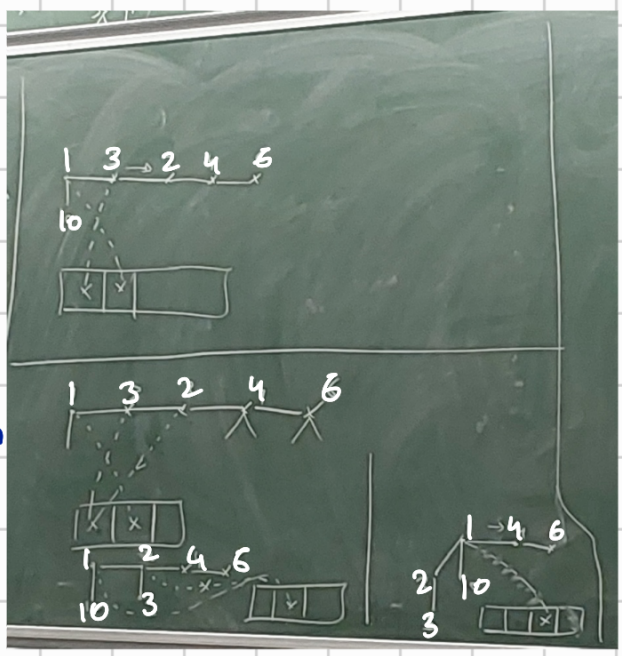
coût amorti : constant + degre(x_{min}) - 1

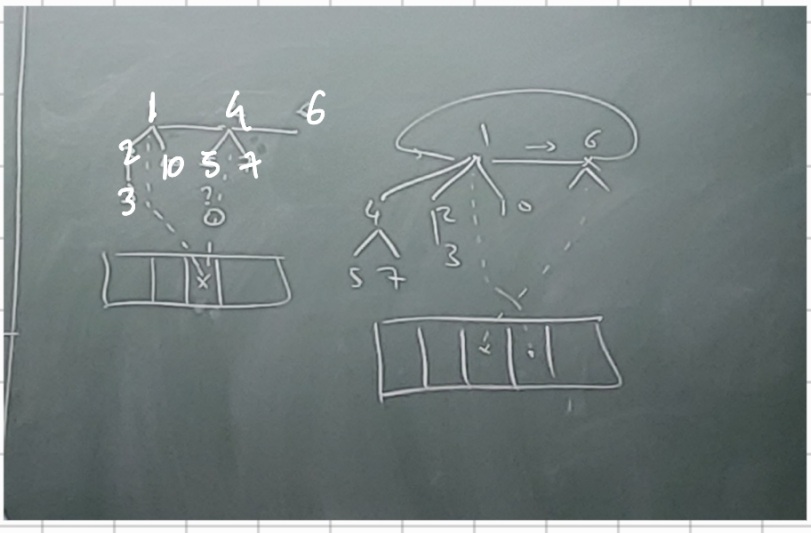
consolidation : ↳ on veut éviter d'avoir des racines avec le même degré (# de fils)



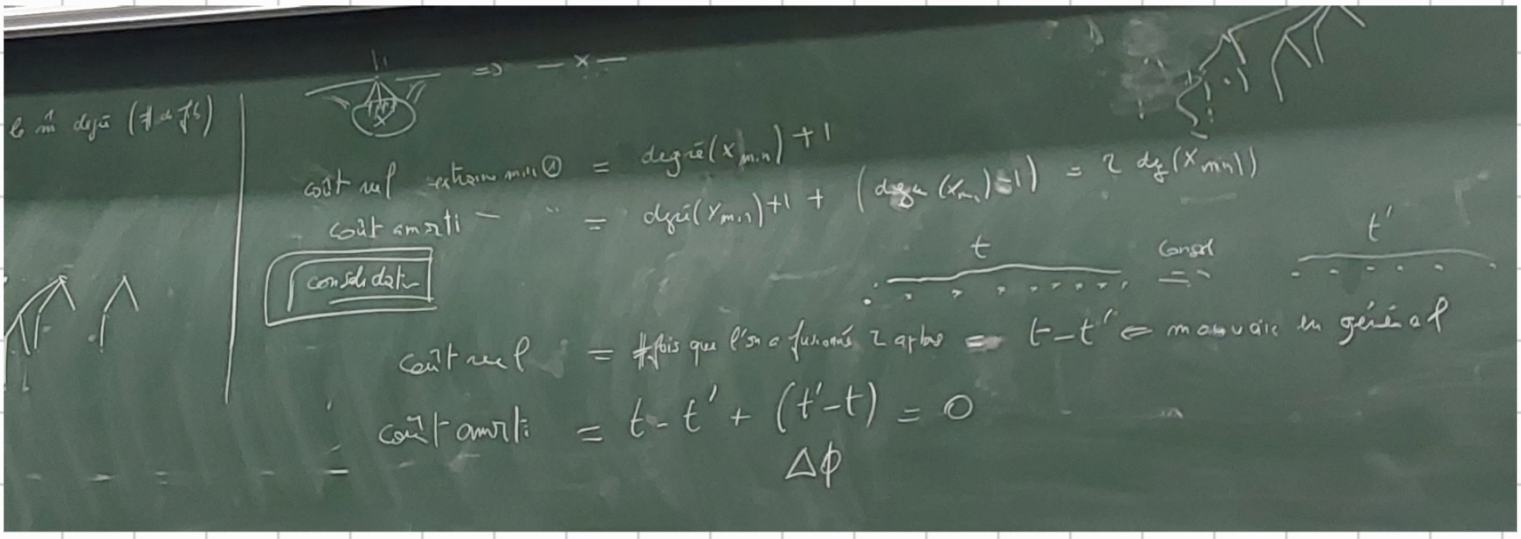




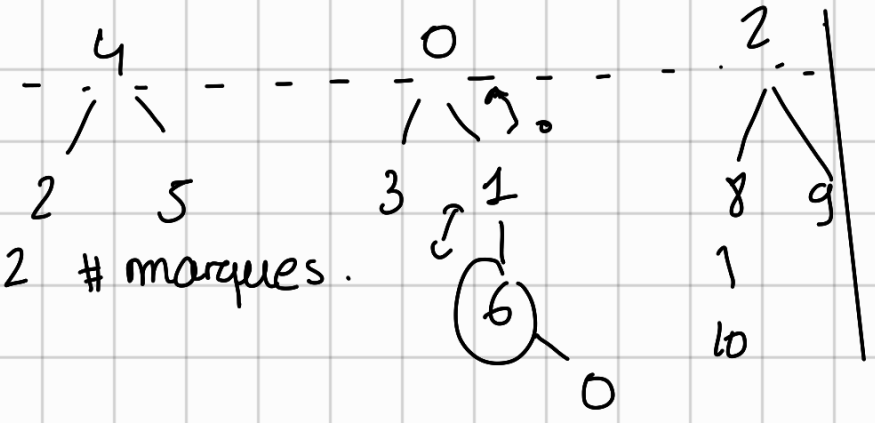




(peut demander sur partiel).

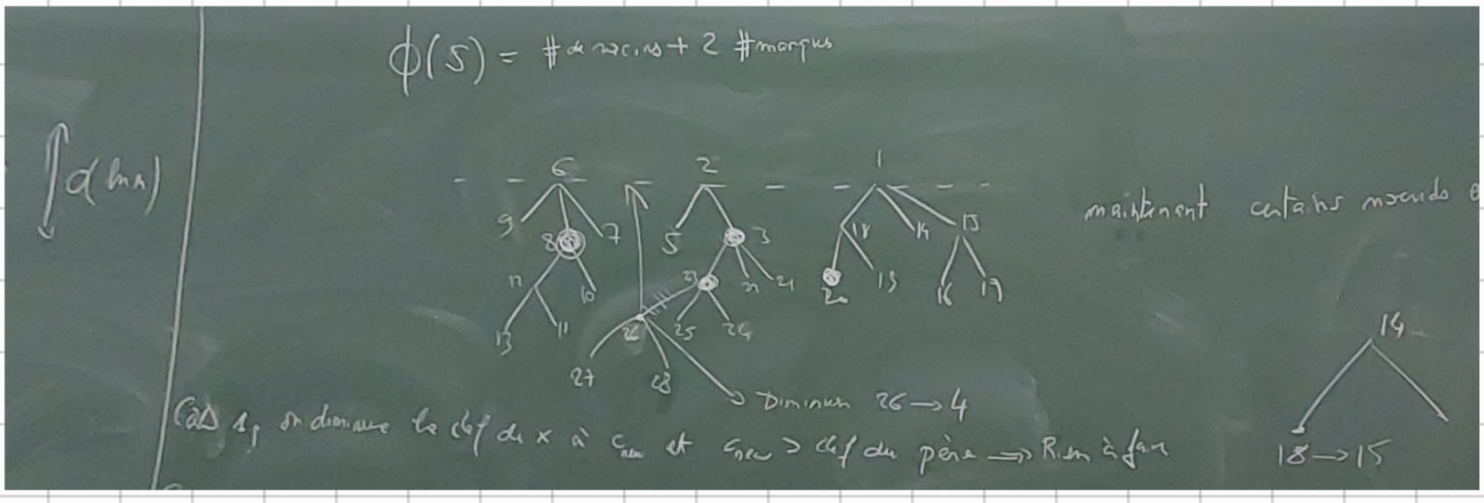


Diminution de clé :



$\phi(S)$ = # de racine + 2 # marques.

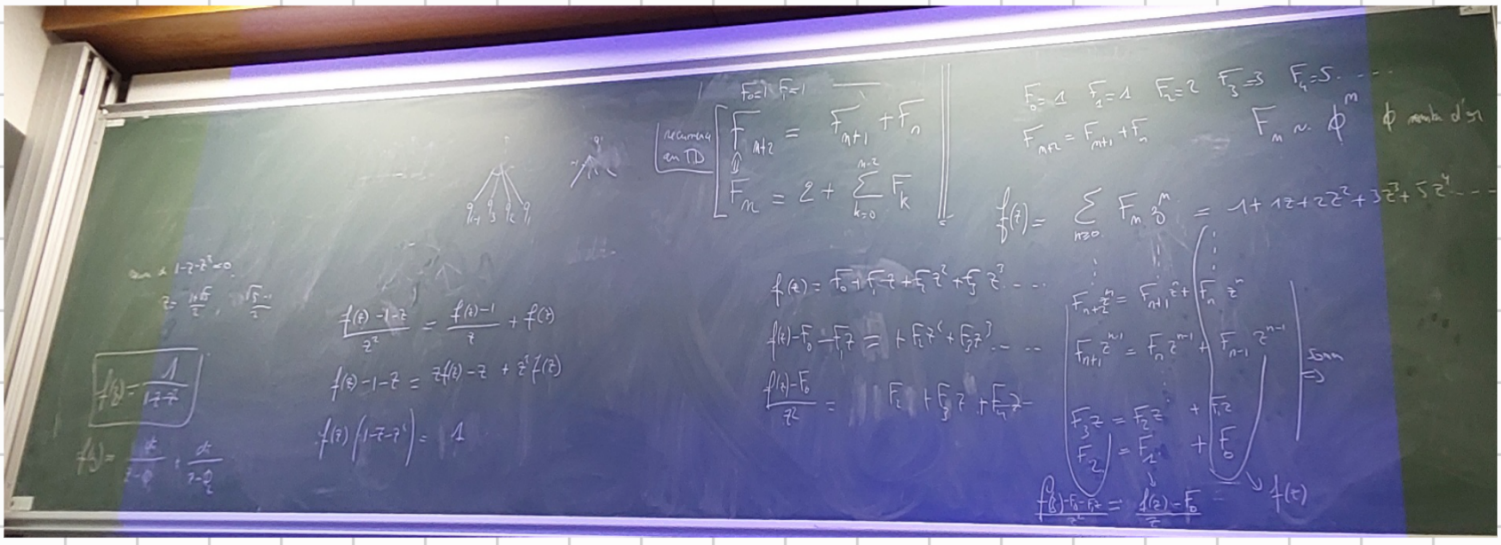
$O(\ln n)$



- **Cas 1 :** on diminue la clef de x à c_{new} et c_{new} > clef du père. ⇒ Rien à faire.

- **Cas 2 :**
  - x_{new} < clé du père ⇒ on met l'arbre de racine c_{new} au niveau des racines.
  - le père n'a pas de marque

- **Cas 3 :**
  - $c_{new} < c_{père}$
  - le père a déjà une marque. (on met des marques → qd on enlève les fils)



$$f(z) = \frac{cte}{1 - \frac{z}{\phi_2}} + \frac{cte}{1 - \frac{z}{\phi_2}}$$

$f(z) = F_0 + F_1 z \dots + F_n z^n$ — F_n croit expo^{+} vite

↳ $F(n) = cte \cdot \phi_1^{-n} + cte \cdot \phi_2^{-n}$.

racine de $z^2 + z - 1 = 0$.

$$\frac{1}{1 - az} = 1 + az + a^2 z^2 + \dots a^n z^n$$